ELEC2134 Laboratory 1

Filters and Frequency Response

Overview

In this laboratory, you will construct a simple filter circuit on your breadboard, and will graph Bode plots of the ideal transfer functions for the circuit. After taking a series of measurements, you will plot the actual frequency response of the circuit, and will compare it against the ideal case.

Aims

- Practise finding the transfer function of circuits using the Laplace transform.
- Construct a Bode plot from measurements taken on a real circuit.
- Determine the cutoff frequencies of filter circuits, analytically and experimentally.

Half-Power Frequencies

The half-power frequency is often also called the *cutoff frequency*, *break frequency*, *corner frequency* or the -3dB point. At the half-power frequencies, the power of the circuit drops to a factor of 1/2 of the maximum and the gain of the circuit drops by a factor of $1/\sqrt{2}$ of the maximum. In terms of decibels, this corresponds to a drop of

$$20\log\left(\frac{1}{\sqrt{2}}\right) = -3.0103\text{dB}$$

hence the name -3dB point. To find the half power frequencies analytically, for a transfer function $G(s)$, determine the maximum gain G_{max} and solve the following equation for ω:

$$|G(s)| = \frac{G_{max}}{\sqrt{2}}.$$

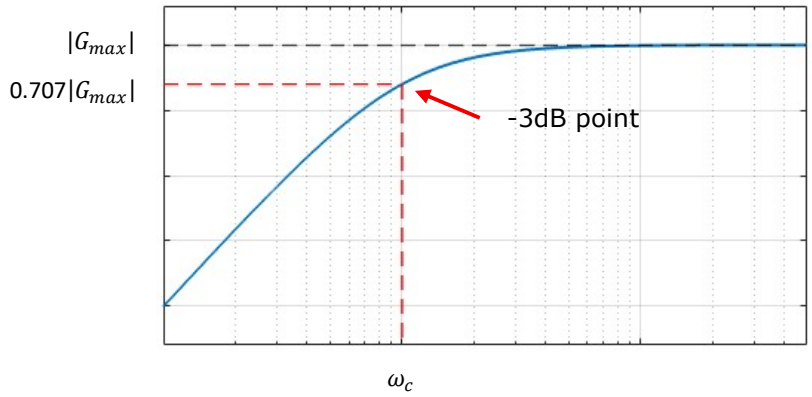


Figure 1.1. $|G_{max}|$ and ω_c indicated on an example Bode plot, with the -3dB point also labelled.

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Prelab Preparation

Section 1. Introduction to LTSpice

Introduction

This initial laboratory exercise aims to familiarise yourself with SPICE (Simulation Program with Integrated Circuit Emphasis), which simulates and verifies circuits. You are expected to draw required circuit schematics, edit simulation commands, and observe simulation results.

Exercises

Download LTSpice simulation software from this website:
https://www.analog.com/en/design-center/design-tools-and-calculators/ltspice-simulator.html

Install LTSpice in your computer and start it. You will see a few drop-down menus and many icons on the top two lines. If you hover your cursor on an icon, you can find its corresponding operation. You can also find these operations from the drop-down menus which show hotkeys and corresponding icons.

The following are the important steps to get simulation results of a circuit:

i. Create a schematic file.
ii. Place components, place power sources and GND – many circuit diagrams do not show GND, but it is important to show GND in schematics for simulation. In this course, only one common GND is needed.
iii. Wire these components.
iv. Create simulation commands (choose from Transient simulation and AC Analysis simulation in this course) and run simulation.
v. Select a node or a component to display voltage/current waveforms in simulation window (or print out) and analyse simulation results.

The circuit in Figure 1.2 corresponds to a voltage divider. Draw the schematic of the circuit shown in Figure 1.2 in LTSpice and simulate it. In the circuit, $v_{in} = 10\,\text{V}$ and $R_1 = R_2 = 10\,\text{k}\Omega$. Show that the output voltage v_{out} in simulation is as you expected, following the steps listed below.

Please follow the appropriate steps for your OS: Windows, Linux or macOS, noting the list of keyboard shortcuts for all OS's are in Table 1.1.

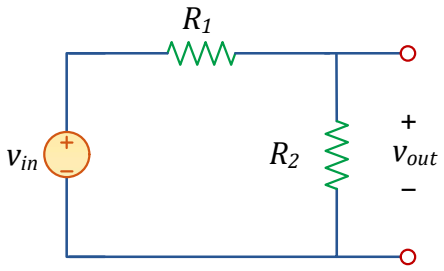


Figure 1.2. Voltage divider circuit.

Instructions for Windows/Linux

1. Create a schematic file. You can find this operation in the 'File' drop-down menu. By clicking this operation, a blank sheet of paper is shown.

2. Place two resistors in the blank area, one by one in the same order so the number will match that in Figure 1.2. You can find 'Resistor' in the 'Edit' drop-down menu. Click it and then click 'Rotate' in the 'Edit' drop-down menu before you place it. Press 'esc' to cancel operations. Right-click the resistor and assign resistance value, 10k. Do the same for the other resistor but no need to rotate.

3. Place power source. Go to 'Component' in the 'Edit' drop-down menu, then select 'voltage'. Right-click on the voltage source symbol and assign DC value, 10V.

4. Place GND. Go to 'Place GND' in the 'Edit' drop-down menu and place the GND symbol below the DC source and resistors.

5. Wiring. Go to 'Draw Wire' in the 'Edit' drop-down menu and click it. Add a wire by clicking the starting terminal, moving to the ending terminal, and clicking again. Connect all the wires as shown in Figure 1.2.

6. Since the voltage at the node between R_1 and R_2 is of interest, we can label the node as 'Vout'. Labelling all the nodes you are interested in will make it easy for you to find and display voltages at these nodes in the simulation window.

   Click 'label Net' in the 'Edit' drop-down menu and type in 'Vout'. Place it as shown in Figure 1.2. Now the schematic is ready for simulation.

7. To set up the simulation command, you can click 'Run' in the 'Simulate' drop-down menu for the first time. Later, if you want to edit it, click 'Edit Simulation Cmd' in the 'Simulate' drop-down menu then click 'Run'. In the pop-up window, choose 'Transient', which is a simulation to calculate current/voltage over time. You can decide simulation duration, say 10ms. So set 'Stop time' to 10ms. You can leave 'Time to start saving data' and 'Maximum Timestep' blank to the default value. After 'OK', place the command (.tran 10m) somewhere.

8. After step 7, a simulation window pops up showing a plot with 0 to 10ms in the horizontal axis. To display the voltage at Vout over time in the vertical axis, click node 'Vout' in the schematic. To see the exact value at a specific time, click 'V(Vout)' on the top of the plot, then a small window pops up to show the value of the horizontal and vertical axes. Drag the vertical line in the plot or press arrow key to see the cursor value changes.

   If you want to display the current through R_2 over time, click the component 'R2' in the schematic.

   Another useful operation is 'Add trace' in the 'Plot Settings' drop-down menu. For example, if you want to plot the instantaneous power consumed by R_2 over time, you can type in 'V(Vout)*I(R2)' in the 'Expression(s) to add' block. V(Vout) is the voltage over R_2 and I(R2) is the current through R_2. It is just like the math operations in the oscilloscope you use in the laboratory.

   To delete a waveform, select 'Delete Traces' in the 'Plot Settings' drop-down menu, then click the name on the top of the plot.

   Other operations can be found in 'Plot Settings' drop-down menu or right-click in the simulation window.

9. After you show that the output voltage is 5V all the time, which is the voltage you would also obtain by analytical calculation, you have successfully completed the task – simulate and verify the DC circuit.

The circuit in Figure 1.2 is a simple DC circuit; however, this course is more about AC circuits. Follow the steps listed below to change the power source from $v_s = 10\,\mathrm{V}$ (DC source) to $v_s = 10\sin(2\pi \cdot 100 \cdot t)\,\mathrm{V}$ (AC source) and investigate how to conduct transient and AC analysis simulation for an AC circuit.

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1. From step 9 in the previous DC circuit, close the simulation window. Right-click on the 10V DC source. Click 'Advanced' and select SINE. Set Amplitude to 10 and Freq to 100. Now the voltage source is $10\sin(2\pi \cdot 100 \cdot t)$ V.

2. If you have not changed any simulation setting, it is still for transient simulation (.tran 20ms). Click 'run' to simulate. Click the node 'Vout' to display the output voltage waveform. You should see a sine wave in one cycle, 10ms, with the peak-to-peak voltage 10V. If you want to see 2 cycles, you can edit the simulation command. Change the 'Stop time' to 20ms.

3. Now we move to AC analysis simulation, which is very common for AC circuits, and you will use a lot in this course. Transient simulation is to show some response (e.g., voltage, current) vs. time, while AC analysis simulation is to show some response vs. frequency.

   Close the simulation window from the previous step. Right-click on the SINE source. Click 'Advanced' and select '(none)'. On the right side of the panel, set 'AC Amplitude' to 1. Now we have done the voltage source setting for AC analysis. In simulation, we will observe how the output voltage changes over different frequencies.

4. To do AC analysis simulation, click 'Edit Simulation Cmd' in the 'Simulate' drop-down menu. Choose 'AC Analysis'. Set 'Type of sweep' to 'Decade'. 'Number of points per decade' is not critical, only affecting waveform resolution. Denser points mean higher resolution at the cost of longer simulation time. You can set it to 10. Start frequency to 1 Hz and Stop frequency to 1 MHz. It means the response will be calculated at each frequency point from 1Hz to 1MHz.

   Now run the simulation. Click the node 'Vout'. You will find the waveform of the output voltage magnitude and phase are displayed. Right-click on the plot left vertical axis, set the 'Representation' to 'Cartesian' from 'Bode'. Now you can see V(Vout) is 0.5V (real part) over all frequencies from 1Hz to 1MHz. This means when the AC source is 1V@1Hz, 1V@2Hz, …, 1V@1MHz, the output voltage magnitude is always 0.5V.

   If you do not understand the meaning, do not worry. You will learn phasor method and frequency response later in the course. Right now, you just need to know how to do AC analysis simulation.

   One important thing to say is that, if you do AC simulation, you must set 'AC Amplitude' and 'AC phase' in the Voltage source setting. If you do transient simulation, you can use any voltage source as needed.

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Instructions for macOS

1. Create a schematic file. You can find this operation in the 'File' drop-down menu (see Figure 1.3). By clicking this operation, a blank sheet of paper is shown.

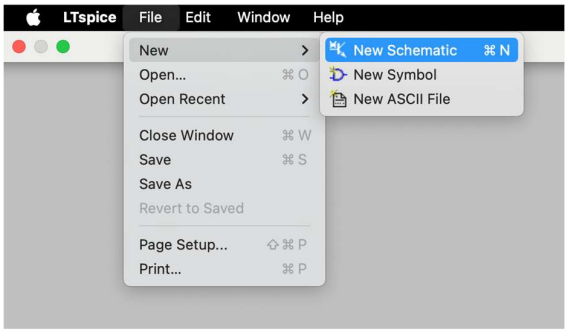


Figure 1.3. Creating a schematic file in LTSpice for macOS.

2. Place two resistors in the blank area, one by one in the same order so the number will match that in Figure 1.2. You can find 'Resistor' by right-clicking an empty space in the schematic, Select Draft -> Component, then select 'Resistor'. **Alternatively, you can press 'R' on the keyboard**. Press Ctrl-R before you place it to rotate the resistor. Press 'esc' to cancel operations. Right-click the resistor and assign resistance value, 10k. Do the same for the other resistor but no need to rotate.

3. Place power source. Go to 'Component' by right-clicking and select Draft -> Component, then select 'voltage'. **Alternatively, press fn-F2 on the keyboard, then select 'voltage'**. Right-click on the voltage source symbol and assign DC value, 10V.

4. Place GND by right-clicking and select Draft -> Net Name, then tick GND (global node 0). Click 'OK'. **Alternatively, press 'G' on the keyboard.** Once GND symbol is obtained, press place the GND symbol below the DC source and resistors.

5. Wiring. Right-click and select Draft -> Wires drop-down menu and click it. **Alternatively, press fn-F3 on the keyboard.** Add a wire by clicking the starting terminal, moving to the ending terminal, and clicking again. Connect all the wires as shown in Figure 1.2.

6. Since the voltage at the node between R_1 and R_2 is of interest, we can label the node as 'Vout'. labelling all the nodes you are interested in will make it easy for you to find and display voltages at these nodes in the simulation window.

   Right-click and select Draft -> Net Name, **or press fn-F4 on the keyboard.** Place it as shown in Figure 1.2. Now the schematic is ready for simulation.

7. To set up the simulation command, right-click -> Draft -> Spice directive, **or press S on the keyboard**, right-click on the blank textbox and click 'Help me edit -> Analysis Cmd' (see Figure 1.4). choose 'Transient', which is a simulation to calculate current/voltage over time. You can decide simulation duration, say 10ms. So set 'Stop time' to 10ms. You can leave 'Time to start saving data' and 'Maximum Timestep' blank to the default value. After 'OK', place the command (.tran 10m) somewhere.

   Then click 'Run' on the top left corner of the windows.

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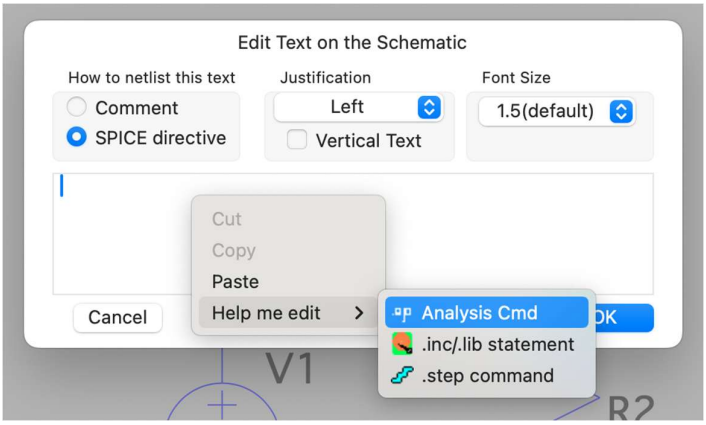


Figure 1.4. Setting up simulation command in LTSpice for macOS.

8. After step 7, a simulation window pops up showing a plot with 0 to 10ms in the horizontal axis. To display the voltage at Vout over time in the vertical axis, click node 'Vout' in the schematic. To see the exact value at a specific time, click 'V(Vout)' on the top of the plot, then a small window pops up to show the value of the horizontal and vertical axes. Drag the vertical line in the plot or press arrow key to see the cursor value changes.

If you want to display the current through $\mathrm{R2}$ over time, click the component 'R2' in the schematic.

Another useful operation is 'Add trace' in the 'Plot Settings' drop-down menu. For example, if you want to plot the instantaneous power consumed by $\mathrm{R2}$ over time, you can type in 'V(Vout)*I(R2)' in the 'Expression(s) to add' block. V(Vout) is the voltage over $\mathrm{R2}$ and I(R2) is the current through $\mathrm{R2}$. It is just like the math operations in the oscilloscope you use in the laboratory.

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1. From step 9 in the previous DC circuit, close the simulation window. Right-click on the 10V DC source. Click 'Advanced' and change the style to SINE. Set DC offset to 0, Amplitude to 10 and Freq to 100. Now the voltage source is $10\sin(2\pi 100t)\ V$

2. If you have not changed any simulation setting, it is still for transient simulation (.tran 20ms). Click 'run' to simulate. Click the node 'Vout' to display the output voltage waveform. You should see a sine wave in one cycle, 10ms, with the peak-to-peak voltage 10V. If you want to see 2 cycles, you can edit the simulation command. Change the 'Stop time' to 20ms.

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4. To do AC analysis simulation, right-click on the simulation command to edit it. Right-click on the textbox and select 'Help me edit -> Analysis Cmd'. Choose 'AC Analysis'. Set 'Type of sweep' to 'Decade'. 'Number of points per decade' is not critical, only affecting waveform resolution. Denser points mean higher resolution at the cost of longer simulation time. You can set it to 10. Start frequency to 1 Hz and Stop frequency to 1 MHz. It means the response will be calculated at each frequency point from 1Hz to 1MHz (1Meg).

Now run the simulation. Click the node 'Vout'. You will find the waveform of the output voltage magnitude and phase are displayed. Right-click on the plot left vertical axis, set the 'Representation' to 'Cartesian' from 'Bode'. Now you can see V(Vout) is 0.5V (real part) over all frequencies from 1Hz to 1MHz. This means when the AC source is 1V@1Hz, 1V@2Hz, …, 1V@1MHz, the output voltage magnitude is always 0.5V.

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One important thing to say is that, if you do AC simulation, you must set 'AC Amplitude' and 'AC phase' in the Voltage source setting. If you do transient simulation, you can use any voltage source as needed.

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List of Keyboard Shortcuts

Table 1.1. List of keyboard shortcuts in LTSpice.

Symbol	Name	Windows / Linux	macOS
	New Schematic	Ctrl N	Cmd N
	Open file	Ctrl O	Cmd O
	Save	Ctrl S	Cmd S
	Rotate	Ctrl R	Cmd R / Ctrl R
	Mirror	Ctrl E	Cmd E / Ctrl E
	Zoom to Rectangle	Ctrl Z	Z
	Zoom Back	Ctrl B	track pad / scroll wheel
	Zoom to Fit	Space bar	Space bar
	Components	F2	fn-F2
	Wire	F3	fn-F3
	label Net	F4	fn-F4
	Cut / Delete	F5 / Delete	fn-F5
	Copy	F6	fn-F6
	Move	F7	fn-F7
	Drag	F8	fn-F8
	Undo	F9	Ctrl Z / Cmd Z / fn-F9
	Redo	Shift-F9	Shift + Ctrl Z / Cmd Z / fn-F9
	GND	G	G
	Resistor	R	R
	Capacitor	C	C
	Inductor	L	L
	Text	T	T
	Spice Directive	S	S

Section 2. Low-Pass Filter Design

1. Derive the transfer function $H(j\omega) = \frac{V_{out}(j\omega)}{V_{in}(j\omega)}$ and an expression for the half-power frequency, ω_c for the passive low-pass filter in Figure 1.5.

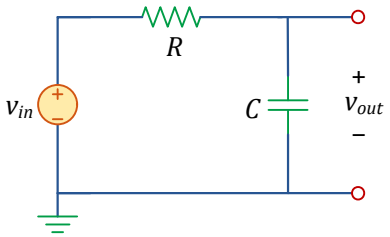


Figure 1.5. RC circuit working as low-pass filter (first-order filter).

2. Choose the component values so that the -3dB point is at 10 krad/s.

3. Simulate the frequency response of the low-pass filter in LTSpice and based on the components values that you chose, validate that the -3dB point is indeed at 10 krad/s.

Please refer to Section 1 of this Prelab, *'Introduction to LTSpice'*, and the guidance video on Moodle for more information about LTSpice.

Once you have completed all the questions from this checkpoint, please submit a marking request here: https://forms.cloud.microsoft/r/9mnwX47BP1.

Checkpoint 1.1	Assessor	Marks	Date

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Lab Exercises

In this lab you will build the low-pass filter shown in Figure 1.5 with the component values that you found in your Prelab so that the -3dB point is at 10 krad/s.

1. Measure the actual values of the components, and record both the ideal and measured values in Table 1.2.

Table 1.2. Component values

Required components	Chosen nominal value/s	DMM or LCR Bridge value/s
R		
C		

2. Determine the actual -3dB point ω_c by measuring the output magnitude, and record your result in Table 1.3. To determine the -3dB point ω_c, you need to first find the frequency response (Bode Plot) of the filter. To do that, you need to find the gain of the filter at different frequencies so that you can plot the response.

3. Apply a 1V peak-to-peak sine wave as an input and measure the peak-to-peak value of the output. The number of measurements you take is up to you, ensure you cover the range from 10Hz to at least 60kHz, and make sure you tabulate all your measurements in Table 1.3 and plot the magnitude response in Figure 1.6.

Table 1.3. Measurements for the low pass filter

Frequency (kHz)											
V_{in} (V)											
V_{out} (V)											
$Gain\ (dB)$											

Frequency (kHz)											
V_{in} (V)											
V_{out} (V)											
$Gain\ (dB)$											

Frequency (kHz)											
V_{in} (V)											
V_{out} (V)											
$Gain\ (dB)$											

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4. Based on the measurements you took in Table 1.3, find the -3dB point and compare it to the ideal -3dB point that you designed for, and plot the frequency response in Figure 1.6.

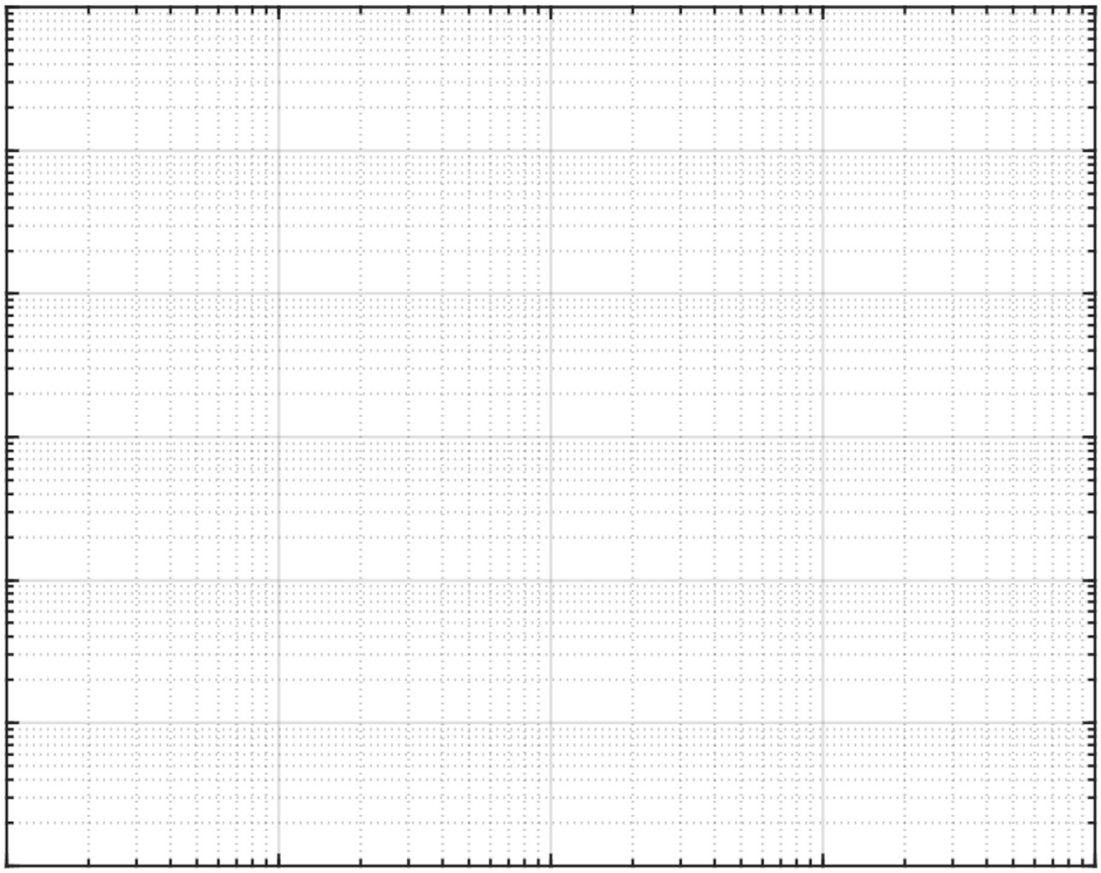


Figure 1.6. Magnitude response of the low-pass filter

Once you have completed all the questions from this checkpoint, please submit a marking request here: https://forms.cloud.microsoft/r/9mnwX47BP1.

Checkpoint 1.2	Assessor	Marks	Date